

PID Control System for Quantum Bit

April 25, 2026

1 echos

Let d be the distance to a flat surface, v be the speed of sound, $\delta(t)$ be the transmitted signal, α be the absorption coefficient. the impulse response of the echo channel is $I(t) = \frac{\delta(t - T)Ae^{-\alpha T}}{\sqrt{d}}, T = 2d/v$. For an unknown environment, an upper boundary of the impulse response of an echo channel is the sum of echos from surfaces at all possible distances $E(t) = \int_0^\infty \frac{\delta(t - T)Ae^{-\alpha T}}{\sqrt{vT/2}} \frac{v}{2} dT = A\sqrt{\frac{v}{2t}}e^{-\alpha t}$. Let d be the distance to another beacon. Then the received response from it is $R(t) = \frac{\delta(t - T)Aexp(-\alpha T)}{\sqrt{d}}, T = d/v$.